



Incident handler's journal

Date: July 23, 2026	Entry: #1
Description	Documenting a cybersecurity incident This incident occurred in the two phases: 1. Detection and Analysis: The scenario outlines how the organization first detected the ransomware incident. For the analysis step, the organization contacted several organizations for technical assistance. 2. Containment, Eradication, and Recovery: The scenario details some steps that the organization took to contain the incident. For example, the company shut down their computer systems. However, since they could not work to eradicate and recover from the incident alone, they contacted several other organizations for assistance.
Tool(s) used	Incident Response Playbook, NIST Security Framework
The 5 W's	• Who: An organized group of unethical hackers • What: A ransomware security incident • Where: At an unspecified health care company • When: Tuesday 9:00 a.m. • Why: The incident happened because unethical hackers were able to access the company's systems using a phishing attack. After gaining access, the attackers launched their ransomware on the company's systems, encrypting critical files. The attackers' motivation appears to be financial because the ransom note they left demanded a large sum of money in exchange for the decryption key.
Additional notes	1. How could the health care company prevent an incident like this from occurring again? 2. Should the company pay the ransom to retrieve the decryption key?

Date: July 25 2026	Entry: #2
Description	Analyzing a packet capture file
Tool(s) used	For this activity, I used Wireshark to analyze a packet capture file. Wireshark is a network protocol analyzer that uses a GUI. Wireshark in cybersecurity allows security analysts to capture and analyze network traffic. This can help in detecting and investigating malicious activity.
The 5 W's	• Who: N/A • What: Data Packet • Where: Windows OS • When: July 25 2026 @ 3:25 pm • Why: To understand the details and provide experience for why to analyze certain aspects of a data packet.
Additional notes	I've never used Wireshark before, so I was excited to begin this exercise and analyze a packet capture file. At first glance, the interface was very overwhelming. I can see why it's such a powerful tool for understanding network traffic.

Date: July 25 2026	Entry: #3
Description	Capturing a data packet
Tool(s) used	tcpdump to capture and analyze network traffic. Tcpdump is a network protocol analyzer that's accessed using the command-line interface. Similar to Wireshark, the value of tcpdump in cybersecurity is that it allows security analysts to capture, filter, and analyze network traffic.
The 5 W's	• Who: N/A • What: tcpdump • Where: N/A • When: N/A • Why: To understand why I might need to use this and gain first-hand

	knowledge of breaking down data packets.
Additional notes	I'm still new to using the command-line interface, although I find it works for me just as well as Wireshark. I took some extra time to look over the commands, after carefully following the instructions, I was able to get through this activity and capture network traffic feeling more confident about what each segment of a data packet entails.

Date: July 27 2026	Entry: #4
Description	Investigate a suspicious file hash
Tool(s) used	For this activity, I used VirusTotal, which is an investigative tool that analyzes files and URLs for malicious content such as viruses, worms, trojans, and more. It's a very helpful tool to use if you want to quickly check if an indicator of compromise like a website or file has been reported as malicious by others in the cybersecurity community. For this activity, I used VirusTotal to analyze a file hash, which was reported as malicious. This incident occurred in the Detection and Analysis phase. The scenario put me in the place of a security analyst at a SOC investigating a suspicious file hash. After the suspicious file was detected by the security systems in place, I had to perform deeper analysis and investigation to determine if the alert signified a real threat.
The 5 W's	• Who: An unknown malicious actor • What: An email sent to an employee contained a malicious file attachment with the SHA-256 file hash of 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b • Where: An employee's computer at a financial services company • When: At 1:20 p.m., an alert was sent to the organization's SOC after the intrusion detection system detected the file • Why: An employee was able to download and execute a malicious file

	attachment via e-mail.
Additional notes	How can this incident be prevented in the future? Should we consider improving security awareness training so that employees are careful with what they click on?

Reflections/Notes:

I really found the activity using tcpdump easier the more I used it. I am new to using the command line, and learning the syntax for a tool like tcpdump was a big learning curve. After reading the directions carefully and understanding the concepts completely was emphasized over just memorizing the codes themselves. I redid the activity and figured out where I had trouble doing any of the tasks. What I learned from this was to carefully read the instructions and work through the process slowly.

After taking this course, my understanding of incident detection and response has definitely evolved. At the beginning of the course, I had some basic understanding of what detection and response entailed, but I didn't fully understand the complexity involved. As I progressed through the course, I learned about the lifecycle of an incident; the importance of plans, processes, and people; and tools used. Overall, I feel that my understanding has changed, and I am equipped with more knowledge and understanding about incident detection and response.

I really enjoyed learning about network traffic analysis and applying what I learned through network protocol analyzer tools. It was my first time learning about network traffic analysis, so it was both challenging and exciting. I found it really fascinating to be able to use tools to capture network traffic and analyze it in real time. I am definitely more interested in learning more about this topic, and I hope to one day become more proficient in using network protocol analyzer tools.
